

ResearchShop Desktop: A Desktop Application for Auditable Gene and Variant Extraction from Biomedical Literature Using Hybrid NER–LLM Architecture

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Abstract

We present ResearchShop Desktop, a free, open-source desktop application that assists biomedical researchers with structured extraction of gene and variant associations from PubMed literature. The system combines PubTator3 Named Entity Recognition, open-access full-text parsing from PubMed Central and Europe PMC, and Google Gemini Flash extraction behind a validation-oriented pipeline. Its safeguards include HGNC gene symbol verification, source-text grounding checks, HGVS-style variant pattern validation, citation cross-referencing using dense sequence matching, and explicit provenance fields for each output row. The application runs on consumer hardware with no server infrastructure; users provide their own Gemini API key and NCBI Entrez email.

The contribution of ResearchShop is a reproducible desktop workflow for paper discovery, open-access full-text acquisition, candidate gene discovery, LLM-assisted structured extraction, validation, and CSV/XLSX export. The software is packaged for macOS, Windows, and Linux, includes an offline test suite for core parser and validation behavior, and persists audit artifacts that help users inspect candidate provenance, gate drops, confidence scores, and extraction failures. ResearchShop is intended as a research acceleration tool; extracted associations require expert review before use in publications, systematic reviews, or clinical contexts.

Keywords: Gene-disease association, Biomedical text mining, Large Language Models, Named Entity Recognition, Hallucination mitigation, Full-text processing

Required Metadata

1. Motivation and Significance

1.1. The Knowledge-Extraction Bottleneck

Genomic medicine faces a fundamental knowledge-extraction bottleneck: while sequencing data is abundant, the interpretation of genetic variants relies on evidence scattered across millions of publications. PubMed indexes more than 40 million citations and abstracts in biomedical and life-science literature [1]. Manual curation processes—such as those underpinning ClinVar and OMIM—cannot keep pace, leading to significant delays in integrating new findings into clinical practice.

This bottleneck has direct consequences. A significant proportion of variants identified through clinical sequencing are

classified as Variants of Uncertain Significance (VUS) due to insufficient published evidence. Even when genetic associations are published, months to years may pass before this information reaches clinical databases. For rare diseases, relevant papers may be scattered across thousands of publications, making comprehensive review impractical without automation.

1.2. Limitations of Existing Approaches

Automated text-mining systems offer a solution but face two critical challenges:

Full-text accessibility. Abstract-only mining misses information that appears in Results, Methods, tables, captions, and supplementary material. PubTator Central reported that concept taggers recovered 56–65% of manual annotations from abstracts but 78–83% from full text, an 18–22 percentage-point increase [2]. This requires robust full-text acquisition strategies.

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Nr.	Code metadata description	Value/Content
C1	Current code version	v1.0.0
C2	Code/repository link	https://github.com/michaluppai/RS_SOFTWAREX
C3	Compute Environment	Local desktop (Electron + Python venv)
C4	Legal Code License	MIT License
C5	Version control	git
C6	Code languages & services	TypeScript, React, Electron, Python 3.11+, BioPython, pandas, google-genai (Gemini API), electron-builder, Tailwind CSS
C7	Compilation requirements	Node.js 18+, Python 3.11+; macOS, Windows, Linux
C8	Developer documentation	https://github.com/michaluppai/RS_SOFTWAREX/blob/main/README.md
C9	Support contact	m.bujniewiczuppai@uw.edu.pl

Table 1: Code metadata

LLM hallucinations.. Large Language Models show promise for information extraction but are prone to generating plausible yet factually incorrect associations [3]. An LLM may generate valid-sounding gene symbols (e.g., “BRCA3”) or extract real gene names from training data that do not appear in the source paper. For biomedical applications, such errors are unacceptable.

Existing tools fall on a spectrum: dedicated NER systems like PubTator Central [2] offer high precision but limited recall and no contextual extraction; LLM-based approaches offer comprehensive extraction but lack validation safeguards. ResearchShop Desktop bridges this gap by combining both approaches in a hybrid architecture with a multi-layer validation framework.

2. Software Description

2.1. System Architecture

ResearchShop Desktop is a cross-platform desktop application built with Electron [4] (TypeScript/React frontend) and Python 3.11+ (extraction pipeline backend). The application runs entirely on the user’s machine with no server infrastructure. Users provide their own Google Gemini API key and NCBI Entrez email; all pipeline secrets are passed to Python through process environment variables rather than command-line arguments.

The frontend communicates with the Python pipeline via an IPC (Inter-Process Communication) bridge. The pipeline process emits structured JSON messages on stdout (PROGRESS:{...}, LOG:{...}, RESULT:{...}) which the Electron main process parses and forwards to the React renderer. Secrets (API key, email) are passed as environment variables to the spawned Python process, never as command-line arguments.

The pipeline follows a seven-stage architecture (Figure 1):

- PubMed Search & Ranking.** Queries NCBI Entrez for papers matching the user’s query, PMID list, or author name. Results are ranked by citation count using iCite [5] (primary) with Semantic Scholar [6] as fallback. Query-mode searches draw up to PUBMED_RELEVANT_COUNT=200

open-access candidates from PubMed before citation ranking trims the pool to the user’s requested top- N ; user-curated PMID lists are taken 1:1 with no widening.

- Gene Relevance Scoring (UI-side).** A TypeScript keyword-and-regex scorer runs in the Electron renderer, scoring each paper for gene relevance before the user submits their selection. High/medium papers are auto-selected; low-relevance papers are hidden by default with a “Show all” toggle. This replaces traditional pipeline-side abstract screening, giving researchers transparent control.
- Full-Text Acquisition.** Retrieves structured full text via PMC Entrez JATS XML (primary) or Europe PMC XML (fallback). The system extracts section structure (abstract, introduction, methods, results, discussion), figure captions and image URLs, and supplementary tables. Only open-access papers are fetched; no paywall bypass is attempted.
- PubTator NER.** Submits PMIDs to NCBI PubTator3 [7] for Named Entity Recognition. Returns high-precision gene symbols and variant annotations. These serve as seed candidates for the LLM stage, forming the *precision floor* of the hybrid architecture.
- LLM Extraction (Gemini Flash).** A two-stage extraction using Google Gemini Flash:
 - Stage A:* Abstract-level gene discovery (fast, cheap screening).
 - Stage B:* Full-text detailed extraction with the user’s custom schema columns, requiring verbatim citations for each claim.

PubTator results are injected as deterministic candidate seeds, constraining the LLM’s hallucination surface. An optional multimodal pass analyses figure images via Gemini Vision.

- Gene Validation.** Multi-source validation against a bundled HGNC database (44,943 genes), HGNC REST API, and MyGene.info. Includes biotype annotation, HGVS variant pattern matching, and citation cross-referencing (Section 2.5).

7. **Orchestration & Output.** A multiprocessing worker pool (2–4 persistent processes, 600 s per-paper timeout) coordinates stages 3–6. Results are deduplicated, ranked by citation count, and written to CSV with full validation metadata.

2.2. Hybrid NER-LLM Architecture

A key design decision is the *hybrid* approach combining two complementary extraction strategies:

- **PubTator3 (NER):** High precision, lower recall. Trained on annotated biomedical literature; genes it identifies are almost certainly real. Cannot extract contextual information (disease associations, functional impact, statistical evidence).
- **Gemini Flash (LLM):** High recall, lower precision. Extracts contextual relationships, statistical evidence, and citations. Prone to hallucination—generating gene symbols from training data that do not appear in the source paper.

The combination operates as follows: PubTator provides the precision floor (safety net), while the LLM extends coverage and adds structured context. Deterministic candidate seeding (`ENABLE_DETERMINISTIC_CANDIDATES`) injects PubTator genes as known-good candidates into the LLM prompt, reducing the hallucination surface. A grounding check then verifies that *every* LLM-extracted gene appears as text in the fetched paper, using the canonical symbol, all HGNC aliases, and the raw label the LLM used.

2.3. Schema-Driven Extraction

Users define custom output columns as natural-language descriptions. For example, researchers might specify:

- **Key Finding:** Main research finding for this gene
- **Statistical Evidence:** p-values, odds ratios, confidence intervals
- **Functional Impact:** Effect on protein function or pathway

The pipeline dynamically constructs LLM prompts incorporating these column definitions, enabling researchers to customise extraction for different contexts (pharmacogenomics, rare diseases, cancer genomics) without code modifications. For each extracted value, the LLM is required to provide a verbatim citation from the source text.

2.4. Desktop Application

The application is packaged as a standalone installer for macOS (.dmg), Windows (.exe via NSIS), and Linux (.AppImage/.deb) using electron-builder. On first launch, the application automatically creates a Python virtual environment and installs pipeline dependencies.

The GUI provides:

- **Onboarding wizard:** API key validation, output directory selection, research-use disclaimer.

- **Query builder:** PubMed search with relevance scoring and paper selection.
- **Live pipeline monitor:** Real-time progress, per-stage breakdown, Gemini API usage tracking, and technical logs for debugging failed runs.
- **Results explorer:** Sortable table with confidence badges (CORROBORATED, MEDIUM, LOW, REVIEW), citation coverage metrics, and CSV/XLSX export.

Security measures include OS-keychain-backed API key encryption (via Electron `safeStorage`), Content Security Policy headers, IPC path traversal validation, sandboxed renderer, and PMID input sanitisation.

2.5. Validation Framework

The validation framework operates at four levels, each targeting different error types:

2.5.1. Gene Symbol Validation

Every extracted gene is verified against authoritative databases in priority order:

1. **Local HGNC database:** A bundled snapshot of 44,943 HGNC entries (19,297 protein-coding) enables offline lookups of official symbols, aliases, and previous symbols.
2. **HGNC REST API:** Fallback for genes not found locally (e.g., recently approved symbols).
3. **MyGene.info:** Second fallback providing additional alias coverage.
4. **Fuzzy matching:** Suggests corrections for near-misses (e.g., “BRAC1” → “BRCA1”).

Biotype information is retained as metadata so researchers can distinguish protein-coding genes, non-coding RNA genes, pseudogenes, and other HGNC locus classes during review. The current release does not reject valid HGNC symbols solely because they are non-protein-coding; this avoids false negatives in RNA-seq, regulatory RNA, and immunogenomics workflows.

2.5.2. Grounding Check

The grounding check verifies that each candidate gene symbol appears as text in the fetched paper. This prevents “contextual hallucinations” where the LLM mentions real genes from its training data that are not discussed in the paper. The check searches for the canonical symbol, all HGNC aliases, and the raw label the LLM used (e.g., “BNP” for NPPB).

2.5.3. Variant Pattern Matching

Extracted variants are validated against 12+ HGVS nomenclature patterns [8]: protein substitution (p.Arg175His), coding variants (c.1234A>G), genomic variants (g.123456A>T), frameshifts, splice sites, indels, copy number states, translocations, and cytogenetic notation. Both three-letter and single-letter amino acid codes are recognised.

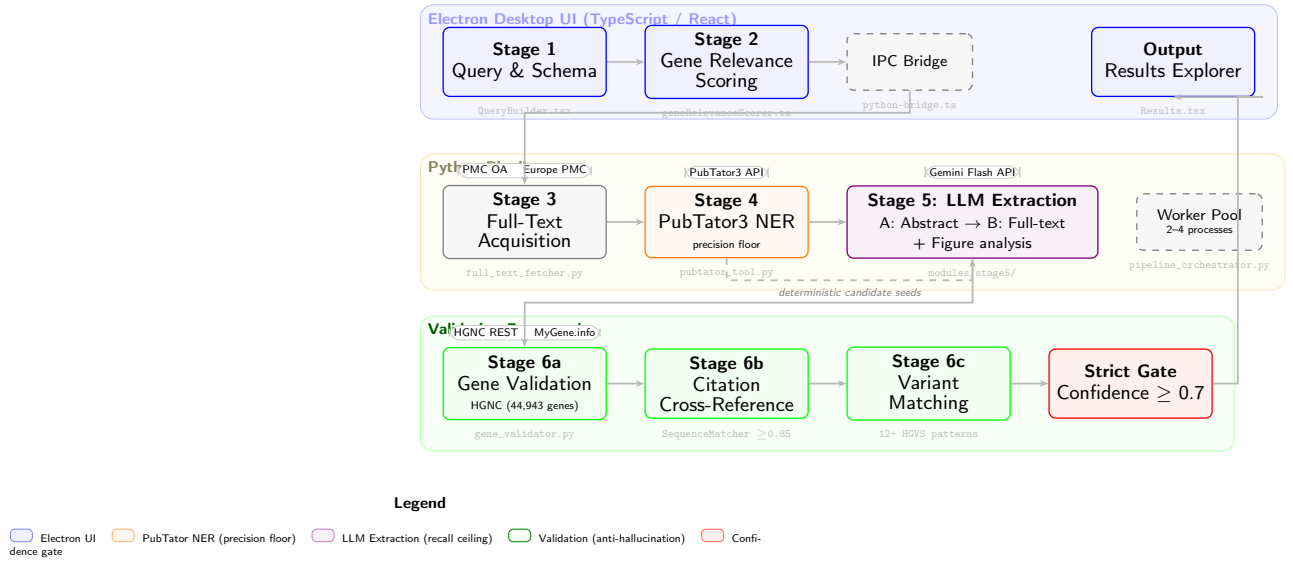


Figure 1: ResearchShop Desktop system architecture. The Electron UI layer (top) handles query input, gene relevance scoring, and results display. The Python pipeline (middle) executes full-text acquisition, PubTator NER, and LLM extraction in a multiprocessing worker pool. The validation framework (bottom) applies gene symbol verification, citation cross-referencing, and a strict confidence gate (≥ 0.7) before output.

2.5.4. Citation Cross-Referencing

For each LLM-provided citation, the system verifies its existence in the paper text using dense sequence matching (`difflib.SequenceMatcher`, threshold ≥ 0.85). Additional checks include:

- **Numerical consistency:** All numbers in the citation must exist in the matched paragraph.
- **Gene context:** The gene symbol or an alias must appear within $\pm 1,500$ characters of the citation match.
- **Encoding normalisation:** Unicode slash variants, LaTeX Greek commands, and ASCII prefixes are normalised symmetrically on both sides before matching.

2.5.5. Confidence Scoring and Strict Gate

Each gene–paper association receives a confidence score:

$$C_{\text{total}} = C_g + C_v \quad (1)$$

where $C_g = 0.7$ if the gene passes HGNC validation (1.0 for gene-only associations) and $C_v = 0.3$ if the variant passes HGVS pattern matching. The strict validation gate (`FINAL_VALIDATION_MIN_CONFIDENCE=0.7`) filters associations below this threshold from the output CSV. This threshold is a medical accuracy decision: lowering it increases recall but also increases false gene associations.

3. Illustrative Examples

3.1. Research Workflow

ResearchShop is designed for a common literature-curation workflow: a researcher starts from a PubMed query or a curated

PMID list, reviews the candidate papers in the desktop interface, selects the papers to process, defines the output columns they need, and receives a spreadsheet of gene–paper associations with provenance and validation metadata. The same backend can also be run from the command line for reproducible scripted analyses.

For example, a researcher studying inflammatory genetics can submit a PMID list and request columns such as *Key Finding*, *Disease Association*, *Statistical Evidence*, and *Conclusion*. The pipeline then fetches open-access full text, parses body text, tables, figures, and supplementary metadata where available, combines PubTator candidates with deterministic HGNC scanning, and asks Gemini to fill the requested schema only for grounded candidate genes. The output includes the extracted fields, citation validation status, NCBI identifiers, candidate source, association type, association group, confidence label, and paper metadata.

3.2. Audit Artifacts

Each run writes more than the final CSV. ResearchShop persists a metadata CSV, XLSX workbook, JSON export, candidate audit artifact, drop-debug artifact, and optional pipeline trace. These artifacts are intentionally verbose: they record which genes were proposed, which gates accepted or rejected them, whether citations were grounded in the source text, and whether output rows originated from LLM extraction, PubTator, deterministic seeding, or fallback paths.

This design is important for scientific review. A researcher can inspect why a gene appeared in the output, why another gene was dropped, and whether the supporting sentence was copied from the paper or generated by the LLM. The application therefore treats extraction as an auditable triage step rather than as an opaque classifier.

3.3. Figure and Table Handling

Many genomics papers encode primary findings in figures, tables, and supplementary files rather than in prose. ResearchShop parses PMC and Europe PMC JATS XML, extracts structured table text, collects figure captions and image URL candidates, and passes table/caption context into evidence validation. When figure analysis is enabled and images are downloadable, Gemini Vision can inspect figures as an optional signal; when images are unavailable, the pipeline degrades to caption and text evidence rather than failing the run.

The parser path uses ResearchShop-owned fetchers for PubMed, PMC, Europe PMC, and citation services. A small adapter uses `pubmed_parser` for selected JATS parsing tasks such as body paragraphs and figure metadata, while ResearchShop keeps its own open-access gates, URL probing, CDN fallback, table extraction, supplementary file handling, PubTator integration, and output schema.

3.4. Reproducibility Checks

The repository includes offline tests for the main scientific risk points: XML parsing, figure metadata extraction, citation grounding, gene validation, F2-isoprostane false-positive suppression, association-type grouping, PubTator/NCBI caching, Stage 5 import compatibility, output writing, and pipeline tracing. These checks do not require a Gemini key or network access and are intended to be run before release builds.

The SoftwareX submission focuses on software architecture, usability, auditability, and reproducibility of the workflow. Historical benchmark scripts and audit records remain in the repository for transparency, but the manuscript does not rely on an expanded external benchmark as a prerequisite for publication.

4. Impact

4.1. Key Contributions

ResearchShop Desktop makes three primary contributions:

1. **Hybrid NER-LLM architecture with validation.** The combination of PubTator3, deterministic HGNC candidate scanning, and Gemini Flash allows the system to identify candidate genes while applying multiple validation gates before output. This addresses a critical gap: neither traditional NER nor unconstrained LLM extraction alone is sufficient for reliable biomedical triage.
2. **Privacy-preserving desktop application.** The tool runs entirely on consumer hardware with no server infrastructure. Patient data, institutional queries, and research directions never leave the user's machine. Users provide their own API key, avoiding shared-infrastructure costs.
3. **Transparent, auditable extraction.** Every output row includes provenance metadata: candidate source, normalisation applied, validation outcome, confidence score, association type, and association group. Separate audit artifacts record which candidate genes appeared before validation and which gates removed them.

4.2. Applications

- **VUS resolution:** Systematically extracts published gene-disease associations to accelerate variant classification.
- **Rare disease research:** Enables comprehensive literature surveys for conditions with scattered, limited publications.
- **Drug target discovery:** Identifies gene-disease associations relevant to therapeutic development across large corpora.
- **Systematic reviews:** Reduces manual effort in evidence synthesis from months to hours.

4.3. Limitations

The following limitations are documented in the software's audit log and should guide interpretation of any ResearchShop output:

- **LLM stochasticity:** Gemini Flash is non-deterministic; the same paper can yield different field wording or candidate ranking across runs. Important reviews should preserve audit artifacts and manually verify primary sources.
- **Open-access only:** Full text is fetched only from open-access PMC and Europe PMC sources. Papers without accessible full text are processed with reduced context, limiting recall and citation grounding.
- **Clinical-vs-molecular ambiguity:** Abbreviations like ESR, AST, and CRP appear as both clinical lab tests and gene symbols. A prompt-based disambiguation clause provides soft mitigation; the corroboration gate provides hard backstop.
- **AI-extracted data requires expert review:** The tool is designed for research acceleration, not clinical decision-making. All outputs should be independently verified before use in publications, systematic reviews, or clinical contexts.
- **Pharmacogenomics and guideline documents:** The pipeline achieves near-zero recall on clinical pharmacogenomics guidelines, where gene-drug associations are embedded in dosing tables and recommendation prose rather than experimental findings sections. The tool is optimised for primary research papers; guideline documents should be processed with domain-specific extraction approaches.
- **Rare disease citation bias:** Citation-count ranking inherently deprioritises rare disease papers, which accumulate fewer citations than cancer genomics or GWAS studies regardless of scientific merit. Researchers studying rare conditions are advised to use direct PMID submission rather than query-based search to ensure relevant low-citation papers are included.
- **Structural variant notation coverage:** The HGVS validation layer recognises common notation patterns (amino acid substitutions, frameshifts, splice-site variants, indels, copy-number variants, and others) but does not capture

all structural rearrangement notations. Variants in non-standard or complex notation pass gene-level validation and appear in output but are not confirmed as HGVS-compliant.

- **Within-paper deduplication:** The first-occurrence deduplication strategy (one row per gene per paper) may collapse distinct variant-phenotype associations into a single row when the same gene is studied in multiple contexts within one paper. Researchers requiring per-variant granularity should treat the output as a starting index and retrieve the primary source for multi-variant genes.
- **Search quality is not a systematic-review guarantee:** PubMed search, citation ranking, and UI-side relevance scoring help users find likely papers, but ResearchShop does not prove recall of all relevant papers for a systematic review question. Users conducting systematic reviews should document their search strategy independently.
- **Batch workflow rather than interactive evidence synthesis:** The current application processes selected papers into spreadsheets. It does not yet provide notebook-style iterative review, claim-level adjudication, or multi-paper consensus inference.

5. Conclusions

We have presented ResearchShop Desktop, an open-source desktop application for auditable gene and variant extraction from PubMed literature. The hybrid NER-LLM architecture combines PubTator3 candidate discovery, deterministic HGNC candidate scanning, Gemini Flash structured extraction, open-access full-text parsing, and a multi-layer validation framework. HGNC verification, source-text grounding, variant pattern matching, citation cross-referencing, confidence scoring, and candidate audit artifacts provide defence-in-depth against common LLM extraction failures.

The software is freely available under the MIT licence as a cross-platform desktop application. It requires no server infrastructure; users provide their own Gemini API key and NCBI Entrez email. The project includes offline tests for parser, validation, tracing, and output behavior, plus release scripts for macOS, Windows, and Linux installers.

Future work. Planned enhancements include: (1) claim-level verification for free-text findings, separating evidence checking for each extracted assertion rather than validating only citation strings; (2) LLM-assisted PubMed query construction with HGNC alias awareness and explicit diffing against the user's original query; (3) ClinicalTrials.gov integration for pharmacogenomics and translational workflows where trial records provide structured gene-drug context; (4) cross-run aggregation and notebook-style result exploration so users can compare repeated runs and adjudicate uncertain rows; (5) expanded variant extraction coverage for complex structural variants and non-HGVS notation; (6) specialised handling of immunoglobulin and HLA nomenclature, whose IMGT and IPD conventions

conflict with HGNC identifiers; and (7) multi-paper relationship inference to aggregate gene-disease evidence across a corpus and identify consensus associations.

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Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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